

RS002 Week 15 — English Worksheet

Topic: Multiple Files — `import` Your Own Data · **Course:** Pokédex App · **Time:** about 45 minutes

Words to know: **import**, **module**, **from import**, **data file**, **main file**

This week your project grows into **two files**. The Pokémon data lives in `pokemon_data.py`, and your game lives in `main.py`, which pulls the data in with `from pokemon_data import pokedex`. This worksheet is all about **reading and thinking about** that code — and then **explaining** the code you wrote in your lesson. Splitting files is how real programs stay organised.

1 · Recall

From memory:

Write a function `battle(p1, p2)` **that returns whichever Pokémon has the higher** `hp + attack` **(return** `None` **on a tie).**

2 · Predict 🧠

Read each block. Write what you think will happen — just from reading.

Two files:

```
# pokemon_data.py
pokedex = [
    {"name": "피카츄", "type": "전기"},
    {"name": "파이리", "type": "불"},
]
```

```
# main.py
from pokemon_data import pokedex
print(len(pokedex))
print(pokedex[0]["name"])
```

When `main.py` runs, what two lines print?


```
# main.py
import random
from pokemon_data import pokedex

print(random.choice(pokedex)["name"])
```

Both `import` styles appear here. What does this line print? Could it change each run?


```
# main.py
from pokemon_data import pokedex
print(pokemon_data)
```

`from ... import pokedex` only brings in `pokedex`. What happens on the `print` line?

3 · Spot the Bug 🐛

Read what each block is **supposed** to do, fix it on paper, then explain the fix.

Bug A — `main.py` should pull in the dex from the data file. Right now nothing is imported, so `pokedex` does not exist.

```
# main.py
print(len(pokedex))
```

Hint: add a `from pokemon_data import pokedex` line at the top.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The data lives in `pokemon_data.py`, but this import names the wrong file.

```
# main.py
from pokemon import pokedex
print(pokedex[0]["name"])
```

Hint: the module name must match the file name (without `.py`).

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — `random.choice` is used but only the data is imported. The program crashes on `random`.

```
# main.py
from pokemon_data import pokedex
print(random.choice(pokedex)["name"])
```

Hint: `random` is a separate module — import it too.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Explain the Code

Read this working two-file Pokédex app. Then answer the questions below — just from reading.

```
# pokemon_data.py – data only
pokedex = [
    {"name": "피카츄", "type": "전기", "hp": 35, "attack": 55},
    {"name": "이상해씨", "type": "풀", "hp": 45, "attack": 49},
    {"name": "파이리", "type": "불", "hp": 39, "attack": 52},
    {"name": "꼬부기", "type": "물", "hp": 44, "attack": 48},
]
```

```

# main.py - the game
import random
from pokemon_data import pokedex

def show_pokemon(p):
    print(f"{p['name']} ({p['type']}) - HP {p['hp']}, ATK {p['attack']}")

def battle(p1, p2):
    score1 = p1["hp"] + p1["attack"]
    score2 = p2["hp"] + p2["attack"]
    if score1 > score2:
        return p1
    elif score2 > score1:
        return p2
    return None

trainer = input("트레이너 이름: ").strip() or "이름 모를 트레이너"
print(f"\n환영합니다, {trainer} 트레이너님!")

print("\n=== 포켓몬 도감 ===")
for i, p in enumerate(pokedex):
    print(f"{i+1}. {p['name']}")

choice = input("\n원하는 포켓몬 이름: ").strip()
my_pokemon = None
for p in pokedex:
    if p["name"] == choice:
        my_pokemon = p
        break

if my_pokemon == None:
    print("도감에 없는 포켓몬이어서 피카츄로 시작합니다.")
    my_pokemon = pokedex[0]

enemy = random.choice(pokedex)
print(f"\n🔴 배틀 시작!")
show_pokemon(my_pokemon)
print("    vs")
show_pokemon(enemy)

winner = battle(my_pokemon, enemy)
if winner == my_pokemon:
    print(f"\n🏆 {trainer} 트레이너님의 승리!")
elif winner == enemy:
    print(f"\n😓 졌습니다... 다시 도전해보세요!")

```

```
else:
    print(f"\n🏴‍☠️ 무승부!")
```

The line `from pokemon_data import pokedex` is in `main.py`. Where does `pokedex` actually come from, and why can `main.py` use it?

Why does this program need `import random` as well as the `from ... import` line? What would break without it?

What does `enemy = random.choice(pokedex)` do, and why might the enemy be different each time you run the program?

Look at the loop that sets `my_pokemon`. What happens if the trainer types a name that is not in the dex?

If you added a new Pokémon to `pokemon_data.py` only, would it show up in `main.py` ? Explain why.

5 · Explain Your Lesson Code 🎥

Today in your lesson you split your app into two files: `pokemon_data.py` and `main.py`. Now **explain the code you wrote**. Record a short video on your phone. You can show it running. Try to use these words: **import, module, from import, data file, main file**.

1. Show `pokemon_data.py` and read out your `pokedex`.
2. Show the `import` line in `main.py` and explain what it brings in.
3. Run `main.py` and walk through one battle, saying what each part does.
4. Explain why splitting the data file and the main file is helpful.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.